

STUDY AND LEARNING CENTRE

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STUDY TIPS

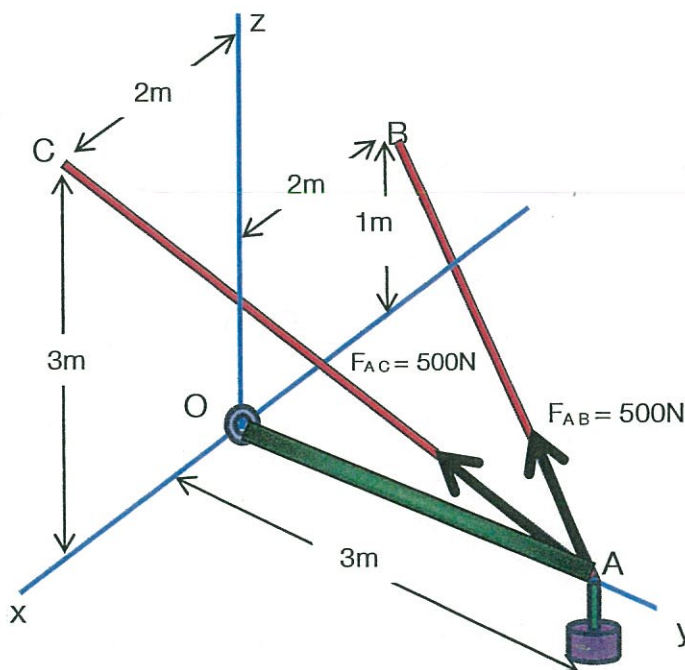


WORKED SOLUTIONS

ENST2.3:

FORCES IN 3 DIMENSIONS

Question Determine (a) the vector forces in the cables F_{AB} and F_{AC} , (b) the resultant vector force F of F_{AB} and F_{AC} , (c) the scalar projection of F along the beam AO , (d) the perpendicular vector component of F to the beam AO , and (e) the angle between the cables AB and AC .



Solution

(a) Position vectors first

$$\vec{OA} = (0, 3, 0) \quad \vec{OB} = (-2, 0, 1) \\ \vec{OC} = (2, 0, 3)$$

$$\vec{AB} = (-2, -3, 1) \quad \text{and}$$

$$\hat{AB} = \frac{1}{\sqrt{14}} (-2, -3, 1)$$

$$\vec{AC} = (2, -3, 3) \quad \text{and}$$

$$\hat{AC} = \frac{1}{\sqrt{22}} (2, -3, 3)$$

$$\vec{F}_{AB} = |\vec{F}_{AB}| \hat{AB} = 500 \frac{1}{\sqrt{14}} (-2, -3, 1) \approx (-267, -401, 134) \text{ N}$$

$$\vec{F}_{AC} = |\vec{F}_{AC}| \hat{AC} = 500 \frac{1}{\sqrt{22}} (2, -3, 3) \approx (213, -320, 320) \text{ N}$$

$$(b) \vec{F} = \vec{F}_{AB} + \vec{F}_{AC} = (-267, -401, 134) + (213, -320, 320) \\ = (-54, -721, 454) \text{ N}$$

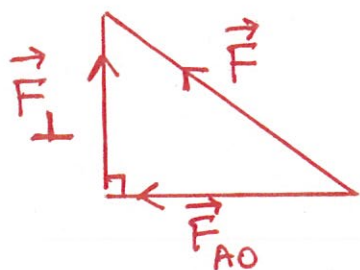
(c) Scalar component of $\vec{F}$ along beam $\vec{AO}$: $\vec{F}_{AO} = \vec{F} \cdot \hat{AO}$

where $\hat{AO} = \frac{\vec{AO}}{|\vec{AO}|} = \frac{(0, -3, 0)}{\sqrt{9}} = (0, -1, 0)$

Hence $\vec{F}_{AO} = (-54, -721, 454) \cdot (0, -1, 0) = 721 \text{ N}$

Note: Since the result is positive $\vec{F}$ has the same sense of direction as $\vec{AO}$

(d) Perpendicular component of $\vec{F}$ to beam $\vec{AO}$



Now $\vec{F}_{AO} = |\vec{F}_{AO}| \hat{AO} = 721 (0, -1, 0)$
 $\vec{F}_{AO} = (0, -721, 0) \text{ N}$

$$\vec{F} = \vec{F}_{\perp} + \vec{F}_{AO} \Rightarrow \vec{F}_{\perp} = \vec{F} - \vec{F}_{AO}$$

$$\therefore \vec{F}_{\perp} = (-54, -721, 454) - (0, -721, 0) = (-54, 0, 454) \text{ N}$$

(e) Angle between 2 vectors is given by dot product

$$\vec{AB} \cdot \vec{AC} = |\vec{AB}| |\vec{AC}| \cos \theta \Rightarrow \theta = \cos^{-1} \left(\frac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}| |\vec{AC}|} \right)$$

$$\therefore \theta = \cos^{-1} \left(\frac{(-2, -3, 1) \cdot (2, -3, 3)}{\sqrt{14} \times \sqrt{22}} \right) = 63^\circ$$